ORF309/MAT380
Discrete Random Variables II: Independence & Conditioning

(pp.31-35)

Ioannis Akrotirianakis

Agenda for today

- *Expectation of* $X + Y$
- Indicator function and Expectations
- *Independence of DRVs*
- *Conditional Expectation*

    - *Completion of Foundations*

Recap from previous Lecture

- **Definition**: If $X : \Omega \to D$ is a **DRV** and $f : D \to R$ is a function, then the **expectation of *X*** is

$$E[f(X)] = \sum_{i \in D} f(i) P\{X = i\}$$

- **Definition**: If $X : \Omega \to D$ is a DRV, then its distribution is given by specifying the probabilities of all its outcomes $(P\{X = i\})_{i \in D}$

- **Definition**: the Joint Distribution of two DRVs $X : \Omega \to D_X$ and $Y : \Omega \to D_Y$ is specified by all probabilities $(P\{X = i, Y = j\})_{i \in D_X, j \in D_Y}$

- **Definition**: Marginal Distribution (when we know the Joint): $P\{X = i\} = \sum_{j \in D_Y} P\{X = i, Y = j\}$

Expectation – Joint and Marginal Distributions

- **Example**: You know the distributions of 2 DRVs X, Y and want to compute $\boldsymbol{E[X+Y]} = ??$

$$\boldsymbol{E[X+Y]} = \sum_{x \in D_X} \sum_{y \in D_Y} (x+y)\, P\{X = x, Y = y\} =$$

$$= \sum_{x \in D_X} \sum_{y \in D_Y} x\, P\{X = x, Y = y\} + \sum_{x \in D_X} \sum_{y \in D_Y} y\, P\{X = x, Y = y\} =$$

$$\sum_{x \in D_X} x \underbrace{\sum_{y \in D_Y} P\{X = x, Y = y\}}_{\substack{\text{Marginal of } X \\ \text{(sum over } y\text{'s)}}} + \sum_{y \in D_Y} y \underbrace{\sum_{x \in D_X} P\{X = x, Y = y\}}_{\substack{\text{Marginal of } Y \\ \text{(sum over } x\text{'s)}}} = \sum_{x \in D_X} x\, P\{X = x\} + \sum_{y \in D_Y} y\, P\{Y = y\} =$$

$$= \boldsymbol{E[X] + E[Y]}$$

Indicator function & Expectation

Indicator function and Expectations

- **Example**:
  - Random Experiment: Flip n **identical** coins. You are interested in the **total number of Heads** that may come up, on average, if you repeat the experiment many times.
  - Define the DRV: $X =$ "the total # of H"
  - **Question**: What is $E[X] =??$
    - Define the events: $A_i =$ "the i-th coin shows H" , $i = 1,2,\dots,n$
    - $P\{A_i\} = p$ and $P\{A_i^c\} = 1 - p$

Indicator function and Expectations

- **Example (cont.)**:

  - Define the indicator function: $1_{A_i} = \begin{cases} 1, if\ A_i = True \\ 0, otherwise \end{cases}$

  - Then we can write $X = 1_{A_1} + 1_{A_2} + \cdots + 1_{A_n}$

  - We have: $E[X] = E\left[1_{A_1} + 1_{A_2} + \cdots + 1_{A_n}\right] = E[1_{A_1}] + E[1_{A_2}] + \cdots + E[1_{A_n}] =$

$$= P\{A_1\} + P\{A_2\} + \cdots + P\{A_n\} = np$$

Indicator function and Expectations

- **Example (cont.)**:
  - **Note**: we did **NOT** have to assume that $A_i \perp\!\!\!\perp A_j, \forall i \neq j$. This means that even if we were not flipping the coins independently (but through a collusion or conspiracy) we would have gotten the same result: $E[X] = np$
    - A possible **collusion** or **conspiracy**: "I flip my coin and then I <u>force everyone</u> to turn their coins to <u>my outcome</u>".
  - **Note**: We will often compute expectations using **<u>tricks</u>** (i.e., using indicator functions) similar with those in this example.

Independence of DRVs

Independence of DRVs

- **Definition**: Two DRVs $X: \Omega \to D_X$ and $Y: \Omega \to D_Y$ are called **independent** ($X \perp\!\!\!\perp Y$) if for every $x \in D_X$ and $y \in D_Y$ we have

$$P\{X = x | Y = y\} = P\{X = x\} \quad OR \quad P\{X = x, Y = y\} = P\{X = x\}P\{Y = y\}$$

- **Remark**: If two DRVs X, Y are **independent** ($X \perp\!\!\!\perp Y$) then their **marginal** distributions → determine their **joint** distribution

- **Note**: We will use the independence of two or more DRVs is used as a modeling assumption (very similar manner as for events)

Independence of DRVs

- **Example**: $\perp\!\!\!\perp$
  - Consider throwing 2 dice and the DRV:

    X ="the outcome of the 1st die" and Y ="the outcome of the 2nd die"

  - The **outcomes** of every die have the <u>same probability distribution</u> and are <u>independent</u>
    - $X \sim \boldsymbol{Unif}(\{1,2,3,4,5,6\})$ and $Y \sim \boldsymbol{Unif}(\{1,2,3,4,5,6\})$
    - We say that X and Y are **i**ndependent **i**dentically **d**istributed DRVs (or X, Y are **i.i.d.**)
  - We want to compute: $\boldsymbol{E}\left[\mathbf{cos}(\boldsymbol{X})\,\boldsymbol{Y}^{\mathbf{2}}\right] = ??$

Independence of DRVs

- **Example (cont.)**:

$$E[\cos(X)Y^2] = \sum_{i=1}^{6}\sum_{j=1}^{6} \cos(i)\, j^2\, P\{X = i, Y = j\} \overset{\perp\!\!\!\perp}{=}$$

$$\sum_{i=1}^{6}\sum_{j=1}^{6} \cos(i)\, j^2\, P\{X = i\}P\{Y = j\} = \sum_{i=1}^{6} \cos(i)\, P\{X = i\} \sum_{j=1}^{6} j^2\ P\{Y = j\} =$$

$$= E[\cos(X)]\, E[Y^2]$$

Independence of DRVs

- **Remark**: Always remember:
  - $E[X+Y] = E[X] + E[Y]$, **for every X, Y**
  - $E[\varphi_1(X)\,\varphi_2(Y)] = E[\varphi_1(X)]\,E[\varphi_2(Y)]$, **<u>only if</u>** $X \perp\!\!\!\perp Y$

Independence of DRVs

- **Remark**: Always remember:

  - $E[X+Y] = E[X] + E[Y]$, **for every X, Y**

  - $E[\varphi_1(X)\,\varphi_2(Y)] = E[\varphi_1(X)]\,E[\varphi_2(Y)]$, **<u>only if</u>** $X \perp\!\!\!\perp Y$

Example:

Consider the DRV: $X =$"the number we see when we throw a fair die".

Consider the DRV": $Y = X$.

Are X, Y independent ???

Independence of DRVs

- **Remark**: Always remember:

  - $E[X+Y] = E[X] + E[Y]$, **for every X, Y**

  - $E[\varphi_1(X)\,\varphi_2(Y)] = E[\varphi_1(X)]\,E[\varphi_2(Y)]$, **<u>only if</u>** $\boldsymbol{X \perp\!\!\!\perp Y}$

Example:

Consider the DRV: $X =$"the number we see when we throw a fair die".

Consider the DRV": $Y = X$.

It is easy to see that X and Y are NOT independent.

We have: $E[X] = 3.5$, $E[Y] = 3.5$, $E[X]E[Y] = 12.25$, $E[XY] = E[X^2] = \frac{91}{6} \approx 15.17$

Generalization of Independence

- **Definition**: The n DRVs $X_1, X_2, \dots, X_n$ are called **independent** iff

$$P\{X_i = x_i \mid X_{j_1} = x_{j_1}, X_{j_2} = x_{j_2}, \dots, X_{j_k} = x_{j_k}\} = P\{X_i = x_i\}$$

OR

$$P\{X_{j_1} = x_{j_1}, X_{j_2} = x_{j_2}, \dots, X_{j_k} = x_{j_k}\} = P\{X_{j_1} = x_{j_1}\}\, P\{X_{j_2} = x_{j_2}\} \dots P\{X_{j_k} = x_{j_k}\}$$

for all $k \leq n$ and $i \neq j_1 \neq j_2 \neq \cdots \neq j_k$

Conditional Expectations

Conditional Expectation

- **Definition**: If $X:\Omega \to D_X$ and $Y:\Omega \to D_Y$ are DRVs and $\varphi: D \to R$ then the conditional expectation of $\varphi(X)$ given $Y = y$ is

$$E[\varphi(X) \mid Y = y] = \sum_{x \in D_x} \varphi(x) P\{X = x | Y = y\}$$

Conditional Expectation

- **Example**: (interpretation and practice for conditional expectation)
  - We throw a die repeatedly
  - We stop throwing the die when it shows 6 for the first time
  - We want to find the expected number of 1s, given that the game ends in k throws

- **Example (cont.):**
  - Let’s define the DRVs:
    - $T =$ “the time (i.e., trial) we throw the first 6” <u>*(it is “game over”)*</u>
    - X = “the number of 1s” <u>*(counts the 1s)*</u>
  - We were asked to compute ???

- **Example (cont.):**
  - Let's define the DRVs:
    - T = "the time (i.e., trial) we throw the first 6" *(it is "game over")*
    - X = "the number of 1s" *(counts the 1s)*
  - We were asked to compute $\boldsymbol{E[X \mid T = k]}$ = ???

- **Example (cont.):**
  - Let's define the DRVs:
    - T = "the time (i.e., trial) we throw the first 6" *(it is "game over")*
    - X = "the number of 1s" *(counts the 1s)*
  - We were asked to compute $\boldsymbol{E[X \mid T = k]}$ = ???
  - We observe that X counts the number of 1s.
    - For this reason, we will define another DRV and the indicator function:
      - Z_i = "the outcome of the i-th trial"

$$\mathbf{1}_{\{Z_i=1\}} = \begin{cases} \mathbf{1}, & \boldsymbol{if\ Z_i = 1} \\ \mathbf{0}, & \boldsymbol{otherwise} \end{cases}$$

- **Example (cont.):**
  - Then $\boldsymbol{X} = \mathbf{1}_{\{Z_1=1\}} + \mathbf{1}_{\{Z_2=1\}} + \cdots + \mathbf{1}_{\{Z_T=1\}}$
  - **Observation**: if the game ends at the k-th throw, we can safely conclude that X counts the # of 1s until the $k-1$ throw

$$\boldsymbol{X} = \sum_{i=1}^{k-1} \mathbf{1}_{\{Z_i=1\}}$$

- **Example (cont.)**
  - Then we can transform the expectation $E[X \mid T = k]$ as follows:

$$\boldsymbol{E}[\boldsymbol{X} \mid \boldsymbol{T} = \boldsymbol{k}] = E\left[\sum_{i=1}^{k-1} 1_{\{Z_i=1\}} \mid T = k\right] = \sum_{i=1}^{k-1} E\left[1_{\{Z_i=1\}} \mid T = k\right]$$

$$= \sum_{i=1}^{k-1} P[Z_i = 1 \mid T = k\} \qquad (1)$$

- **Example (cont.):**
  - We need to compute: $P[Z_i = 1 \mid T = k\}$ for all $i = 1, \dots, k-1$
  - We will compute it for $\boldsymbol{i = 1}$ and the others are computed similarly

How can I ***equivalently*** write the event $\{T = k\}$??

- **Example (cont.):**
  - We need to compute: $P[Z_i = 1 \mid T = k\}$ for all $i = 1, \dots, k-1$
  - We will compute it for $\boldsymbol{i = 1}$ and the others are computed similarly
  - **Observation**: $\{T = k\} = \{Z_1 \neq 6, Z_2 \neq 6, \dots, Z_{k-1} \neq 6, Z_k = 6\}$

$$P\{Z_1 = 1 \mid T = k\} = P\{Z_1 = 1 \mid Z_1 \neq 6, Z_2 \neq 6, \dots, Z_{k-1} \neq 6, Z_k = 6\} =$$

$$\frac{P\{\boldsymbol{Z_1 = 1}, \boldsymbol{Z_1} \neq \boldsymbol{6}, Z_2 \neq 6, \dots, Z_{k-1} \neq 6, Z_k = 6\}}{P\{Z_1 \neq 6, Z_2 \neq 6, \dots, Z_{k-1} \neq 6, Z_k = 6\}} = \quad \rightarrow$$

$$= \frac{P\{\boldsymbol{Z_1 = 1, Z_1 \neq 6}\}P\{Z_2 \neq 6\} \ldots P\{Z_{k-1} \neq 6\}P\{Z_k = 6\}}{P\{Z_1 \neq 6\}P\{Z_2 \neq 6\} \ldots P\{Z_{k-1} \neq 6\}P\{Z_k = 6\}} =$$

$$\frac{P\{\boldsymbol{Z_1 = 1, Z_1 \neq 6}\}}{P\{Z_1 \neq 6\}} = \frac{P\{\boldsymbol{Z_1 = 1}\}}{P\{Z_1 \neq 6\}} = \frac{1/6}{5/6} = \frac{1}{5}$$

Working similarly, we can find that : $\boldsymbol{P\{Z_i = 1 \mid T = k\} = 1/5}$**, for all** $\boldsymbol{i = 1, \ldots, k-1}$

Hence: $\quad \boldsymbol{E[X \mid T = k] = \sum_{i=1}^{k-1} P\{Z_i = 1 \mid T = k\} = (k-1)/5}$

$\blacksquare$

Practice exercises/examples

(may not be covered during lectures but they are offered for your practice)

- **Exercise - Calculating the Expected Number of Snowfall Records**: We start recording annual snowfall from this year.
- Define the indicator variable: X_j as follows:
    - $X_j = 1$ if year j is a "record-snow year" (snowfall is larger than all previous years); $X_j = 0$, otherwise.
- Assume that the amount of snowfall in different years are independent and follow the same distribution (i.i.d.).
    - **Q1.** if we look at the first j years, is there any reason to believe that the highest amount of snowfall is more likely to happen in year 1 or year j? <u>Hint</u>: think of it as a game, where there are j players each equally likely to win a game;
    - **Q2.** Let S_n be the total number of records in n years. How can you express S_n in terms of the indicator variables X_j
    - **Q3.** Show that the expected number of records is $E[S_n] = \sum_{j=1}^{n} 1/j$
    - **Q4.** the summation $\sum_{j=1}^{n} 1/j$ is quite famous in the mathematical world. Do your research to find its behavior when $n \to \infty$ (**suggestion**: use internet search instead of GenAI tools, because you will find much more information about it)